INTRODUCTION TO WAN TECHNOLOGIES

Introduction

A Wide Area Network (WAN) connects computers and local networks across large geographical distances. Organisations use WANs to connect branch offices, cloud platforms and data centres. Multiprotocol Label Switching (MPLS) is a technology used by service providers to forward traffic efficiently through a provider network.

Task 1: WAN vs. LAN Comparison

A Local Area Network (LAN) covers a limited area such as a home, office or campus. A WAN connects networks over larger distances, often using a telecom or Internet service provider.

Feature	LAN	WAN	Daily-life example
Coverage	Small area: room, home, office or campus.	Large area: cities, states or countries.	Home Wi-Fi is a LAN; accessing IRCTC servers from Ahmedabad uses a WAN/Internet path.
Ownership	Usually managed by a person or one organisation.	Often uses infrastructure operated by telecom or Internet providers.	An office manages its own LAN; a company may buy WAN connectivity from an ISP.
Speed and latency	Usually high speed and low latency over short distances.	Performance depends on distance, provider links and network congestion.	Streaming a video on a home Wi-Fi LAN is local access; the video service is reached over a WAN.
Cost and setup	Usually less expensive and simpler to set up.	Usually more costly and requires provider links or Internet connectivity.	A home router creates a LAN; a business connecting Ahmedabad and Bangalore offices needs WAN services.

Note: Instagram, Zomato, IRCTC and Flipkart are online services. A user's device connects through a local LAN (such as Wi-Fi), while traffic to the service's remote systems travels across the Internet, which is a WAN.

Task 2: WAN Network Diagram

The diagram shows an Ahmedabad user accessing Flipkart's Bangalore data centre. The user's device first connects to the Ahmedabad LAN. A router then forwards the traffic through a WAN/Internet provider network. At the destination, the Bangalore data-centre network delivers the traffic to the application servers.

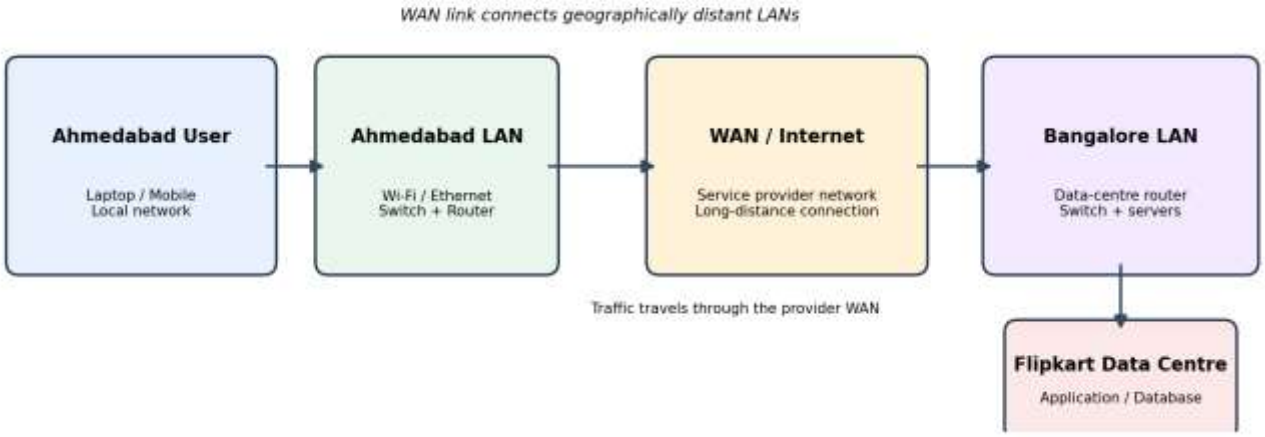


Figure 1: Example of LAN-to-WAN-to-LAN communication.

Task 3: What is MPLS? (5–7 lines)

1. Multiprotocol Label Switching (MPLS) is a packet-forwarding technology commonly used in service-provider and enterprise WAN networks.
2. Instead of making every forwarding decision only by checking the destination IP address at each router, MPLS adds a short label to a packet.
3. At the network edge, an ingress router classifies the traffic and attaches an appropriate label.
4. Routers inside the MPLS network, called Label Switch Routers (LSRs), use the label to forward packets along a Label Switched Path (LSP).
5. The labels can be swapped at each hop, and the egress router removes the label before forwarding the packet onward.
6. MPLS can support traffic engineering, quality-of-service policies and separation of customer networks using technologies such as MPLS VPNs.
7. These features help providers manage traffic predictably and efficiently, although actual performance depends on network design and service configuration.

Task 4: Observations in a Pre-configured MPLS Network

The following are two typical behaviours to look for in the supplied Packet Tracer or GNS3 topology. The exact commands and output depend on the file and the router platform.

Observation 1: Label-based forwarding

At the ingress edge router, an IP packet is classified and an MPLS label is added. Inside the provider network, core routers forward the packet by examining the label rather than performing a fresh destination-IP route lookup for every hop. A label may be swapped for a different outgoing label at the next router. This is different from ordinary IP forwarding, where routers use the destination IP address and their IP routing table at each hop.

Observation 2: Label Switched Path (LSP)

Packets belonging to the same forwarding class can follow a pre-established Label Switched Path (LSP) through the MPLS core. The path is set up using a label-distribution or signalling mechanism, depending on the network design. The egress router removes the MPLS label and forwards the original packet toward its destination. In standard IP routing, each router independently selects the next hop using its routing table; MPLS core forwarding follows the label-switched path.

Suggested checks (if supported by the device): use `show mpls ldp neighbor`, `show mpls forwarding-table`, `show mpls interfaces`, and `show ip route`. Compare the MPLS forwarding table with the IP routing table. In GNS3, command availability depends on the router image and IOS version. Packet Tracer may not support all MPLS features, so use the supplied topology's available commands.

Task 5: Three Benefits of MPLS for High-Traffic Companies

1. Traffic engineering and efficient paths

MPLS can steer selected traffic over paths chosen by the network operator instead of relying only on the shortest IP route. For a service such as YouTube, this can help distribute heavy video traffic across available links and reduce congestion on busy paths.

2. Quality of Service (QoS) support

MPLS networks can classify and prioritise different traffic types. For BookMyShow, time-sensitive application traffic can be given suitable priority during a high-demand ticket sale, helping the service maintain more predictable performance when the network is busy. QoS cannot create extra bandwidth, but it helps manage how available bandwidth is used.

3. VPN and network separation

MPLS VPN services can keep the traffic of different customers or company sites logically separated across a shared provider backbone. A company with offices in Ahmedabad, Mumbai and Bangalore can connect its sites over a managed WAN without building a separate physical backbone for every site.

Conclusion

LANs connect devices within a limited location, while WANs connect networks over long distances. MPLS improves the way provider networks forward and manage traffic by using labels and Label Switched Paths. Its traffic-engineering, QoS and VPN capabilities make it useful for organisations that need reliable connectivity between branches, users and data centres.

Practical Submission Note

Task 4 is written as the expected general behaviour of an MPLS-enabled network. Before submitting, compare these observations with the actual Packet Tracer/GNS3 file provided by your instructor and add the router names, label values or screenshots visible in that topology.